

# ADHITYA M R

Backend Engineer | AI/LLM Application Developer

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## Education

**Bharathidasan University**

Aug 2022 – Apr 2026

*B.Tech in Computer Science Engineering, CGPA: 7.44*

## Experience

**AI Product Development Intern**

Jan 2026 – Jun 2026

*KData Inc.*

*Toronto, Canada (Remote, flexible hours)*

- Implemented Python modules for the BRD parsing layer, extracting structured API endpoint definitions and test assertions from unstructured requirement documents
- Drove LLM selection for the I-QA test generation pipeline by evaluating Gemma4, Qwen3, and Llama3 against expected test structures, iterating on system prompts until output accuracy met production requirements
- Tested the end-to-end pipeline across varied BRD input formats, identifying and resolving parsing edge cases to improve pipeline robustness
- Maintained on-premises environment configurations keeping all LLM inference and database operations within the private network boundary, with zero external data egress

**AI Engineer, Lead Developer**

Dec 2025 – Apr 2026

*Capstone Project – RAG Chatbot*

- Designed, built, and deployed a FastAPI backend from scratch over institutional document data, live on campus servers serving 3,500+ students and staff with no external LLM dependency
- Built a hybrid retrieval layer (FAISS dense vectors + BM25 keyword search, fused via Reciprocal Rank Fusion) outperforming dense-only search in internal evaluation
- Built an incremental ingestion system with SQLite-backed file metadata tracking, re-indexing only new/modified PDFs and keeping ingestion time constant (2.5s/doc) as the corpus grows
- Implemented real-time token streaming via FastAPI Server-Sent Events (P95 first-token latency: 450ms); hardened with LLM circuit breakers, anti-hallucination prompting, and XSS sanitization on model output

## Projects

**FitNova Sales-Call Intelligence** | *FastAPI, SQLAlchemy, SQLite, Groq/Whisper*

Jul 2026

- Built an end-to-end sales-call intelligence pipeline for a take-home technical assessment: ingestion, transcription, diarization, LLM-based quality scoring, and a role-based dashboard for Directors, Team Leads, and Advisors
- Hardened the pipeline against real-world failure modes: SHA-256 upload idempotency, exponential-backoff API retries, and regex-based PII redaction
- Designed a 5-dimension weighted quality rubric and an 8-tag, 3-severity issue taxonomy enforced via strict JSON schema validation, with a post-processing filter that drops any tag outside the taxonomy to eliminate hallucinated flags
- Modeled a source-agnostic ingestion layer (Org → Teams → Advisors) supporting new teams/advisors with zero schema changes, plus a human-feedback loop for advisors to contest flags for Team Lead review

## Technical Skills

**Languages:** Python, Java, SQL, Bash

**Backend & APIs:** FastAPI, REST API Design, Pydantic, asyncio, WebSockets/SSE, Spring Boot

**AI/LLM Engineering:** RAG Pipelines, Agentic Workflows (LangGraph, LangChain), Hybrid Search (BM25 + Dense + RRF Fusion), Prompt Engineering, LLM Evaluation (Ragas), Vector Databases (FAISS, ChromaDB), Local LLM Deployment (Ollama)

**Databases:** PostgreSQL, SQLite, Redis

**Reliability & Data Integrity:** Idempotency (hash-based dedup), Retry/Backoff Strategies, Schema Validation, PII Redaction

**DevOps & Cloud:** Docker, AWS, GCP, GitHub Actions, CI/CD, Linux, Git, pytest, Postman

**Certifications:** Anthropic Academy (Claude & Prompt Engineering), LangChain Academy (LangChain & LangGraph), Google Cloud (Generative AI Fundamentals)